

Phu Nguyen

Going into this lab, I honestly thought EDA was just looking at the data real quick before the "real" work started. But after going through it, I see it's more like being a detective, where you look for clues before you build anything. Just using `.info()` and `.describe()` showed me stuff I wouldn't have thought about, like how Age and Cabin had a lot of missing values, and that way more people died than survived, which could be a problem for a model later.

The plots were the part that helped me the most. The countplots made it really easy to see that 1st class passengers and women survived way more than 3rd class passengers and men. The age histogram also showed a bump of young kids who survived, which fit the whole "women and children first" idea. For my own experiments, I was a little surprised that Cherbourg had the highest survival rate, since I figured the port someone boarded at wouldn't matter. But I think it's probably because more of those passengers were in 1st class, so class was likely the real reason and not the port. The fare boxplot kind of backed that up, since survivors paid more on average, and I noticed a few really high fares that stood out as outliers.

I also learned that a plot shows things a number like the mean can't, like how spread out the data is and where the outliers are. I don't want to read too much into one chart though, because a pattern in a graph doesn't mean one thing caused the other. Overall this lab made EDA feel a lot more useful than I expected, and I can see why you'd do it before modeling.

Cyndy Ndong

In this lab, I worked on exploration data analysis tasks using Python libraries like pandas, matplotlib, and seaborn. The goal was to inspect the Titanic dataset, check for missing values, and look for obvious patterns before building models. This helped me understand the difference between supervised and unsupervised learning. Since our dataset included the Survived column as an answer key, this was a supervised learning task. If we didn't have that column and were

just grouping passengers by age or class, it would be unsupervised. It showed me why checking and cleaning data first is so important.

Looking at the graphs showed clear trends. Passengers in first class and those who paid higher fares had much better survival rates than those in third class. Also, people who boarded at Cherbourg survived at a higher rate than those from Southampton. Seeing this made it easy to see why picking the right features matters so a model learns real patterns instead of random noise.

I hit a few syntax errors when running code in Colab because some lines got squished together when pasting. I fixed them by splitting the code onto separate lines and re-running the cells. After that, all the charts were rendered correctly, and I was able to finish the lab.